

Project Report: Catch the Apple Game (using python – turtle)

Catch the Apple Game

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Declaration

We hereby declare that the project work entitled “Catch the Apple Game” submitted to Lachoo Memorial College, BCA B1 IV Semester, is our own original work and has not been submitted earlier for award of any degree, diploma or certificate. (Signatures)

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1. INTRODUCTION

This project is a simple computer graphics game developed using Python and the Turtle module. The objective of the game is to catch the falling apple using a basket controlled by the user. The game demonstrates basic concepts of animation, user input, and collision detection in computer graphics.

2. OBJECTIVE

The main objectives of this project are:

- To understand the basics of Turtle graphics in Python
 - To create a simple interactive game
 - To implement keyboard controls
 - To apply collision detection logic
 - To develop a scoring system
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3. SOFTWARE REQUIREMENTS

- Python 3.x
 - Turtle module (built-in)
 - Any IDE (IDLE / VS Code / PyCharm)
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4. SYSTEM REQUIREMENTS

- Minimum 4GB RAM
 - Any operating system (Windows/Linux/Mac)
 - Basic computer with keyboard
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5. MODULE DESCRIPTION

1. Screen Setup:

Used to create the game window with specific size and background color.

2. Basket (Player):

A rectangular shape controlled by left and right arrow keys to catch the apple.

3. Falling Object (Apple):

A circular object that falls from the top at a constant speed.

4. Movement Control:

Keyboard inputs are used to move the basket left and right.

5. Collision Detection:

Checks if the apple touches the basket.

6. Score System:

Score increases when the apple is successfully caught.

6. WORKING OF THE PROJECT

- The game starts with a screen and a basket at the bottom.
 - An apple starts falling from a random position at the top.
 - The user moves the basket using arrow keys.
 - If the basket catches the apple, the score increases.
 - If the apple is missed, it resets and falls again.
 - The game continues infinitely.
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7. ALGORITHM

1. Start the program
 2. Initialize screen and score
 3. Create basket and apple
 4. Take keyboard input for movement
 5. Move apple downward continuously
 6. Check collision between basket and apple
 7. If collision occurs, increase score
 8. Reset apple position
 9. Repeat steps until program ends
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8. OUTPUT

The output is a graphical window where the user controls a basket to catch falling apples and score increases accordingly.

9. ADVANTAGES

- Easy to understand and implement
 - Improves programming and logic building skills
 - Demonstrates real-time animation
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10. LIMITATIONS

- Simple graphics only
 - No advanced levels or difficulty
 - No sound effects
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11. FUTURE ENHANCEMENT

- Add multiple apples
 - Increase speed with levels
 - Add sound effects
 - Introduce game over condition
 - Add lives system
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12. CONCLUSION

This project successfully demonstrates the use of Python Turtle graphics to create a simple interactive game. It helps in understanding basic game development concepts like animation, user input, and collision detection.

THANK YOU!